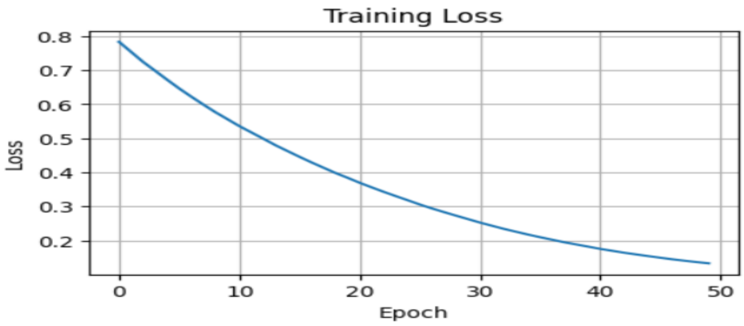
18TH Experiment

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import make_classification

from sklearn.metrics import accuracy_score

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Dense


# Generate linearly separable dataset

X, y = make_classification(

    n_samples=200,

    n_features=2,

    n_redundant=0,

    n_informative=2,

    n_clusters_per_class=1,

    class_sep=2,

    random_state=42

)


# Create Neural Network

model = Sequential([

    Dense(1, activation='linear', input_shape=(2,))

])


# Compile Model

model.compile(

    optimizer='adam',

    loss='mse',

    metrics=['accuracy']

)
```

```python
# Train Model
history = model.fit(
    X, y,
    epochs=50,
    verbose=0
)

# Plot Training Loss
plt.figure(figsize=(5,3))
plt.plot(history.history['loss'])
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(True)
plt.show()

# Prediction
pred = model.predict(X)
pred_class = (pred > 0.5).astype(int)

# Accuracy
acc = accuracy_score(y, pred_class)

print("Accuracy:", round(acc*100,2), "%")

# Plot Dataset
plt.figure(figsize=(5,4))
plt.scatter(X[:,0], X[:,1], c=y)
plt.title("Two-Class Dataset")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
```

plt.show()




```
/7 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step
.ccuracy: 95.0 %
```